Class 13: Pathway Analysis from RNA-Seq Results

Abigail Perez

Overview

Pathway analysis maps differentially expressed genes to known biological pathways, helping us interpret large gene lists and gain insight into the molecular mechanisms underlying a condition.

Section 1. Differential Expression Analysis

Setup

```
library(DESeq2)
```

```
Loading required package: S4Vectors

Loading required package: stats4

Loading required package: BiocGenerics

Loading required package: generics


Attaching package: 'generics'
```

```
The following objects are masked from 'package:base':

    as.difftime, as.factor, as.ordered, intersect, is.element, setdiff,
    setequal, union


Attaching package: 'BiocGenerics'

The following objects are masked from 'package:stats':

    IQR, mad, sd, var, xtabs

The following objects are masked from 'package:base':

    anyDuplicated, aperm, append, as.data.frame, basename, cbind,
    colnames, dirname, do.call, duplicated, eval, evalq, Filter, Find,
    get, grep, grepl, is.unsorted, lapply, Map, mapply, match, mget,
    order, paste, pmax, pmax.int, pmin, pmin.int, Position, rank,
    rbind, Reduce, rownames, sapply, saveRDS, table, tapply, unique,
    unsplit, which.max, which.min


Attaching package: 'S4Vectors'

The following object is masked from 'package:utils':

    findMatches

The following objects are masked from 'package:base':

    expand.grid, I, unname

Loading required package: IRanges

Loading required package: GenomicRanges

Loading required package: Seqinfo

Loading required package: SummarizedExperiment
```

```
Loading required package: MatrixGenerics

Loading required package: matrixStats


Attaching package: 'MatrixGenerics'

The following objects are masked from 'package:matrixStats':

    colAlls, colAnyNAs, colAnys, colAvgsPerRowSet, colCollapse,
    colCounts, colCummaxs, colCummins, colCumprods, colCumsums,
    colDiffs, colIQRDiffs, colIQRs, colLogSumExps, colMadDiffs,
    colMads, colMaxs, colMeans2, colMedians, colMins, colOrderStats,
    colProds, colQuantiles, colRanges, colRanks, colSdDiffs, colSds,
    colSums2, colTabulates, colVarDiffs, colVars, colWeightedMads,
    colWeightedMeans, colWeightedMedians, colWeightedSds,
    colWeightedVars, rowAlls, rowAnyNAs, rowAnys, rowAvgsPerColSet,
    rowCollapse, rowCounts, rowCummaxs, rowCummins, rowCumprods,
    rowCumsums, rowDiffs, rowIQRDiffs, rowIQRs, rowLogSumExps,
    rowMadDiffs, rowMads, rowMaxs, rowMeans2, rowMedians, rowMins,
    rowOrderStats, rowProds, rowQuantiles, rowRanges, rowRanks,
    rowSdDiffs, rowSds, rowSums2, rowTabulates, rowVarDiffs, rowVars,
    rowWeightedMads, rowWeightedMeans, rowWeightedMedians,
    rowWeightedSds, rowWeightedVars

Loading required package: Biobase

Welcome to Bioconductor

    Vignettes contain introductory material; view with
    'browseVignettes()'. To cite Bioconductor, see
    'citation("Biobase")', and for packages 'citation("pkgname")'.


Attaching package: 'Biobase'

The following object is masked from 'package:MatrixGenerics':

    rowMedians
```

```
The following objects are masked from 'package:matrixStats':

    anyMissing, rowMedians
```

```r
library(ggplot2)
```

Load Data

```r
colData <- read.csv("GSE37704_metadata.csv", row.names = 1)
head(colData)
```

```
              condition
SRR493366 control_sirna
SRR493367 control_sirna
SRR493368 control_sirna
SRR493369      hoxa1_kd
SRR493370      hoxa1_kd
SRR493371      hoxa1_kd
```

```r
countData <- read.csv("GSE37704_featurecounts.csv", row.names = 1)
head(countData)
```

```
                length SRR493366 SRR493367 SRR493368 SRR493369 SRR493370
ENSG00000186092    918         0         0         0         0         0
ENSG00000279928    718         0         0         0         0         0
ENSG00000279457   1982        23        28        29        29        28
ENSG00000278566    939         0         0         0         0         0
ENSG00000273547    939         0         0         0         0         0
ENSG00000187634   3214       124       123       205       207       212
                SRR493371
ENSG00000186092         0
ENSG00000279928         0
ENSG00000279457        46
ENSG00000278566         0
ENSG00000273547         0
ENSG00000187634       258
```

Q. Complete the code below to remove the troublesome first column from countData

The first column (`length`) contains gene length information, not counts, so we remove it with `-1`.

```
countData <- as.matrix(countData[, -1])
head(countData)
```

```
                 SRR493366 SRR493367 SRR493368 SRR493369 SRR493370 SRR493371
ENSG00000186092         0         0         0         0         0         0
ENSG00000279928         0         0         0         0         0         0
ENSG00000279457        23        28        29        29        28        46
ENSG00000278566         0         0         0         0         0         0
ENSG00000273547         0         0         0         0         0         0
ENSG00000187634       124       123       205       207       212       258
```

Q. Complete the code below to filter countData to exclude genes (i.e. rows) where we have 0 read count across all samples (i.e. columns).

`rowSums()` adds counts across all samples per gene. Genes with a sum of zero have no data and are removed.

```
countData <- countData[rowSums(countData) > 1, ]
head(countData)
```

```
                 SRR493366 SRR493367 SRR493368 SRR493369 SRR493370 SRR493371
ENSG00000279457        23        28        29        29        28        46
ENSG00000187634       124       123       205       207       212       258
ENSG00000188976      1637      1831      2383      1226      1326      1504
ENSG00000187961       120       153       180       236       255       357
ENSG00000187583        24        48        65        44        48        64
ENSG00000187642         4         9        16        14        16        16
```

Running DESeq2

```
dds <- DESeqDataSetFromMatrix(countData = countData,
                              colData = colData,
                              design = ~condition)
```

```
Warning in DESeqDataSet(se, design = design, ignoreRank): some variables in
design formula are characters, converting to factors
```

```r
dds <- DESeq(dds)
```

```
estimating size factors

estimating dispersions

gene-wise dispersion estimates

mean-dispersion relationship

final dispersion estimates

fitting model and testing
```

```r
dds
```

```
class: DESeqDataSet
dim: 15280 6
metadata(1): version
assays(4): counts mu H cooks
rownames(15280): ENSG00000279457 ENSG00000187634 ... ENSG00000276345
  ENSG00000271254
rowData names(22): baseMean baseVar ... deviance maxCooks
colnames(6): SRR493366 SRR493367 ... SRR493370 SRR493371
colData names(2): condition sizeFactor
```

```r
res <- results(dds)
```

Q. Call the summary() function on your results to get a sense of how many genes are up or down-regulated at the default 0.1 p-value cutoff.

```r
summary(res)
```

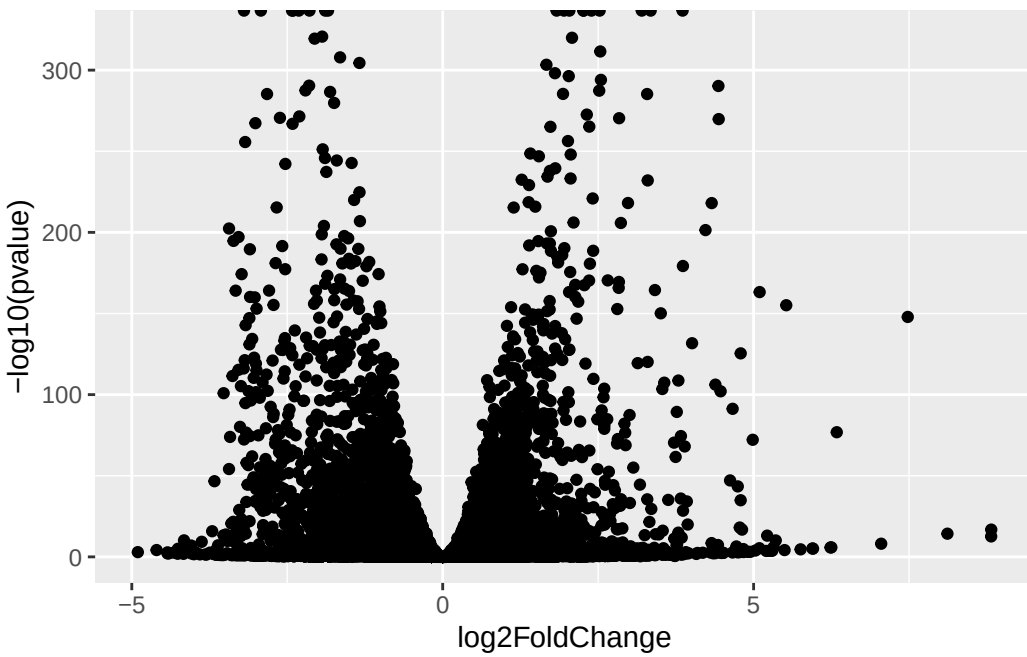
```
out of 15280 with nonzero total read count
adjusted p-value < 0.1
LFC > 0 (up)       : 4351, 28%
LFC < 0 (down)     : 4399, 29%
outliers [1]       : 0, 0%
low counts [2]     : 590, 3.9%
(mean count < 1)
[1] see 'cooksCutoff' argument of ?results
[2] see 'independentFiltering' argument of ?results
```

At the default adjusted p-value cutoff of 0.1, there are 4,351 upregulated genes (28%) and 4,399 downregulated genes (29%).

Volcano Plot

```r
library(ggplot2)

ggplot(as.data.frame(res)) +
  aes(x = log2FoldChange, y = -log10(pvalue)) +
  geom_point()
```

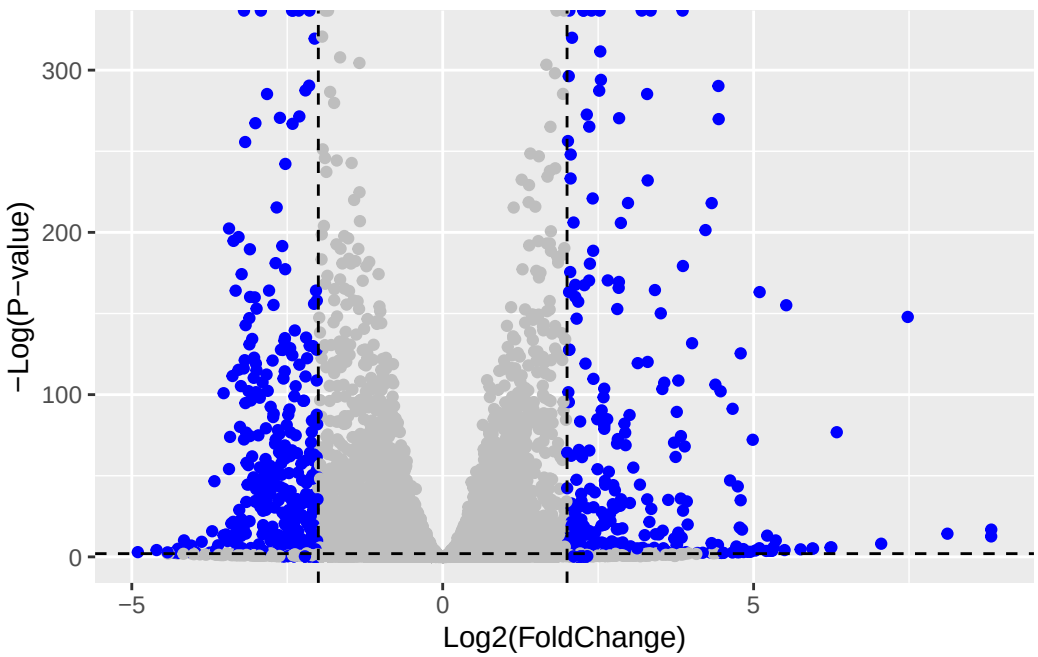


Q. Improve this plot by completing the below code, which adds color, axis labels and cutoff lines:

Blue points have an absolute log2 fold change greater than 2 and an adjusted p-value below 0.01, indicating the most significantly differentially expressed genes.

```
# Color vector
mycols <- rep("gray", nrow(res))
mycols[abs(res$log2FoldChange) > 2] <- "blue"
mycols[res$padj > 0.01] <- "gray"

ggplot(as.data.frame(res)) +
  aes(x = log2FoldChange, y = -log10(pvalue)) +
  geom_point(color = mycols) +
  xlab("Log2(FoldChange)") +
  ylab("-Log(P-value)") +
  geom_vline(xintercept = c(-2, 2), linetype = "dashed") +
  geom_hline(yintercept = -log10(0.01), linetype = "dashed")
```



Adding gene annotation

Q. Use the mapIDs() function multiple times to add SYMBOL, ENTREZID and GENENAME annotation to our results by completing the code below.

```
library(AnnotationDbi)
library(org.Hs.eg.db)
```

```
res$symbol <- mapIds(org.Hs.eg.db,
                     keys = row.names(res),
                     keytype = "ENSEMBL",
                     column = "SYMBOL",
                     multiVals = "first")
```

```
'select()' returned 1:many mapping between keys and columns
```

```r
res$entrez <- mapIds(org.Hs.eg.db,
                     keys = row.names(res),
                     keytype = "ENSEMBL",
                     column = "ENTREZID",
                     multiVals = "first")
```

```
'select()' returned 1:many mapping between keys and columns
```

```r
res$name <- mapIds(org.Hs.eg.db,
                   keys = row.names(res),
                   keytype = "ENSEMBL",
                   column = "GENENAME",
                   multiVals = "first")
```

```
'select()' returned 1:many mapping between keys and columns
```

```r
head(res, 10)
```

```
log2 fold change (MLE): condition hoxa1 kd vs control sirna
Wald test p-value: condition hoxa1 kd vs control sirna
DataFrame with 10 rows and 9 columns
                 baseMean log2FoldChange     lfcSE       stat      pvalue
                <numeric>      <numeric> <numeric>  <numeric>   <numeric>
ENSG00000279457   29.9136      0.1792571 0.3248215   0.551863 5.81042e-01
ENSG00000187634  183.2296      0.4264571 0.1402658   3.040350 2.36304e-03
ENSG00000188976 1651.1881     -0.6927205 0.0548465 -12.630156 1.43993e-36
ENSG00000187961  209.6379      0.7297556 0.1318599   5.534326 3.12428e-08
ENSG00000187583   47.2551      0.0405765 0.2718928   0.149237 8.81366e-01
ENSG00000187642   11.9798      0.5428105 0.5215598   1.040744 2.97994e-01
ENSG00000188290  108.9221      2.0570638 0.1969053  10.446970 1.51281e-25
ENSG00000187608  350.7169      0.2573837 0.1027266   2.505522 1.22271e-02
ENSG00000188157 9128.4394      0.3899088 0.0467164   8.346302 7.04333e-17
ENSG00000131591  156.4791      0.1965923 0.1456109   1.350121 1.76977e-01
                      padj      symbol      entrez                   name
                 <numeric> <character> <character>            <character>
ENSG00000279457 6.85033e-01          NA          NA                     NA
ENSG00000187634 5.14038e-03      SAMD11      148398 sterile alpha motif ..
ENSG00000188976 1.75978e-35       NOC2L       26155 NOC2 like nucleolar ..
ENSG00000187961 1.13044e-07      KLHL17      339451 kelch like family me..
ENSG00000187583 9.19159e-01     PLEKHN1       84069 pleckstrin homology ..
```

```
ENSG00000187642 4.02066e-01       PERM1       84808 PPARGC1 and ESRR ind..
ENSG00000188290 1.30113e-24        HES4       57801 hes family bHLH tran..
ENSG00000187608 2.36679e-02       ISG15        9636 ISG15 ubiquitin like..
ENSG00000188157 4.20596e-16        AGRN      375790                  agrin
ENSG00000131591 2.60893e-01    C1orf159       54991 chromosome 1 open re..
```

Q. Finally for this section let's reorder these results by adjusted p-value and save them to a CSV file in your current project directory.

```
res <- res[order(res$pvalue), ]
write.csv(res, file = "deseq_results.csv")
```

Section 2. Pathway Analysis

KEGG pathways

```
library(pathview)
```

```
##############################################################################
Pathview is an open source software package distributed under GNU General
Public License version 3 (GPLv3). Details of GPLv3 is available at
http://www.gnu.org/licenses/gpl-3.0.html. Particullary, users are required to
formally cite the original Pathview paper (not just mention it) in publications
or products. For details, do citation("pathview") within R.

The pathview downloads and uses KEGG data. Non-academic uses may require a KEGG
license agreement (details at http://www.kegg.jp/kegg/legal.html).
##############################################################################
```

```
library(gage)
```

```r
library(gageData)

data(kegg.sets.hs)
data(sigmet.idx.hs)

# Focus on signaling and metabolic pathways only
kegg.sets.hs <- kegg.sets.hs[sigmet.idx.hs]
head(kegg.sets.hs, 3)
```

```
$`hsa00232 Caffeine metabolism`
[1] "10"   "1544" "1548" "1549" "1553" "7498" "9"

$`hsa00983 Drug metabolism - other enzymes`
 [1] "10"     "1066"   "10720"  "10941"  "151531" "1548"   "1549"   "1551"
 [9] "1553"   "1576"   "1577"   "1806"   "1807"   "1890"   "221223" "2990"
[17] "3251"   "3614"   "3615"   "3704"   "51733"  "54490"  "54575"  "54576"
[25] "54577"  "54578"  "54579"  "54600"  "54657"  "54658"  "54659"  "54963"
[33] "574537" "64816"  "7083"   "7084"   "7172"   "7363"   "7364"   "7365"
[41] "7366"   "7367"   "7371"   "7372"   "7378"   "7498"   "79799"  "83549"
[49] "8824"   "8833"   "9"      "978"

$`hsa00230 Purine metabolism`
  [1] "100"    "10201"  "10606"  "10621"  "10622"  "10623"  "107"    "10714"
  [9] "108"    "10846"  "109"    "111"    "11128"  "11164"  "112"    "113"
 [17] "114"    "115"    "122481" "122622" "124583" "132"    "158"    "159"
 [25] "1633"   "171568" "1716"   "196883" "203"    "204"    "205"    "221823"
 [33] "2272"   "22978"  "23649"  "246721" "25885"  "2618"   "26289"  "270"
 [41] "271"    "27115"  "272"    "2766"   "2977"   "2982"   "2983"   "2984"
 [49] "2986"   "2987"   "29922"  "3000"   "30833"  "30834"  "318"    "3251"
 [57] "353"    "3614"   "3615"   "3704"   "377841" "471"    "4830"   "4831"
 [65] "4832"   "4833"   "4860"   "4881"   "4882"   "4907"   "50484"  "50940"
 [73] "51082"  "51251"  "51292"  "5136"   "5137"   "5138"   "5139"   "5140"
 [81] "5141"   "5142"   "5143"   "5144"   "5145"   "5146"   "5147"   "5148"
 [89] "5149"   "5150"   "5151"   "5152"   "5153"   "5158"   "5167"   "5169"
 [97] "51728"  "5198"   "5236"   "5313"   "5315"   "53343"  "54107"  "5422"
[105] "5424"   "5425"   "5426"   "5427"   "5430"   "5431"   "5432"   "5433"
[113] "5434"   "5435"   "5436"   "5437"   "5438"   "5439"   "5440"   "5441"
[121] "5471"   "548644" "55276"  "5557"   "5558"   "55703"  "55811"  "55821"
[129] "5631"   "5634"   "56655"  "56953"  "56985"  "57804"  "58497"  "6240"
[137] "6241"   "64425"  "646625" "654364" "661"    "7498"   "8382"   "84172"
[145] "84265"  "84284"  "84618"  "8622"   "8654"   "87178"  "8833"   "9060"
[153] "9061"   "93034"  "953"    "9533"   "954"    "955"    "956"    "957"
```

```
[161] "9583"   "9615"
```

```r
foldchanges <- res$log2FoldChange
names(foldchanges) <- res$entrez
head(foldchanges)
```

```
     1266     54855      1465      2034      2150      6659
-2.422719  3.201955 -2.313738 -1.888019  3.344508  2.392288
```

```r
keggres <- gage(foldchanges, gsets = kegg.sets.hs)
```

```r
attributes(keggres)
```

```
$names
[1] "greater" "less"    "stats"
```

Top Downregulated Pathways

```r
# Look at top downregulated pathways
head(keggres$less)
```

```
                                          p.geomean stat.mean        p.val
hsa04110 Cell cycle                    1.003993e-05 -4.353454 1.003993e-05
hsa03030 DNA replication               8.909558e-05 -3.968611 8.909558e-05
hsa03013 RNA transport                 1.470985e-03 -3.007794 1.470985e-03
hsa04114 Oocyte meiosis                1.946905e-03 -2.921710 1.946905e-03
hsa03440 Homologous recombination      2.941989e-03 -2.868141 2.941989e-03
hsa00010 Glycolysis / Gluconeogenesis 6.059196e-03 -2.558327 6.059196e-03
                                            q.val set.size         exp1
hsa04110 Cell cycle                    0.001606390      120 1.003993e-05
hsa03030 DNA replication               0.007127646       36 8.909558e-05
hsa03013 RNA transport                 0.077876200      143 1.470985e-03
hsa04114 Oocyte meiosis                0.077876200       99 1.946905e-03
hsa03440 Homologous recombination      0.094143663       28 2.941989e-03
hsa00010 Glycolysis / Gluconeogenesis 0.161578552       48 6.059196e-03
```

The top downregulated pathway is Cell cycle (hsa04110), which makes sense since HOXA1 loss disrupts cell cycle progression.

Pathview for Cell Cycle

```r
pathview(gene.data = foldchanges, pathway.id = "hsa04110")
```

```
'select()' returned 1:1 mapping between keys and columns

Info: Working in directory /Users/abbyperez/Library/Mobile Documents/com~apple~CloudDocs/BIM

Info: Writing image file hsa04110.pathview.png
```

```r
# PDF version
pathview(gene.data = foldchanges, pathway.id = "hsa04110", kegg.native = FALSE)
```

```
'select()' returned 1:1 mapping between keys and columns

Warning: reconcile groups sharing member nodes!

     [,1] [,2]
[1,] "9"  "300"
[2,] "9"  "306"

Info: Working in directory /Users/abbyperez/Library/Mobile Documents/com~apple~CloudDocs/BIM

Info: Writing image file hsa04110.pathview.pdf
```

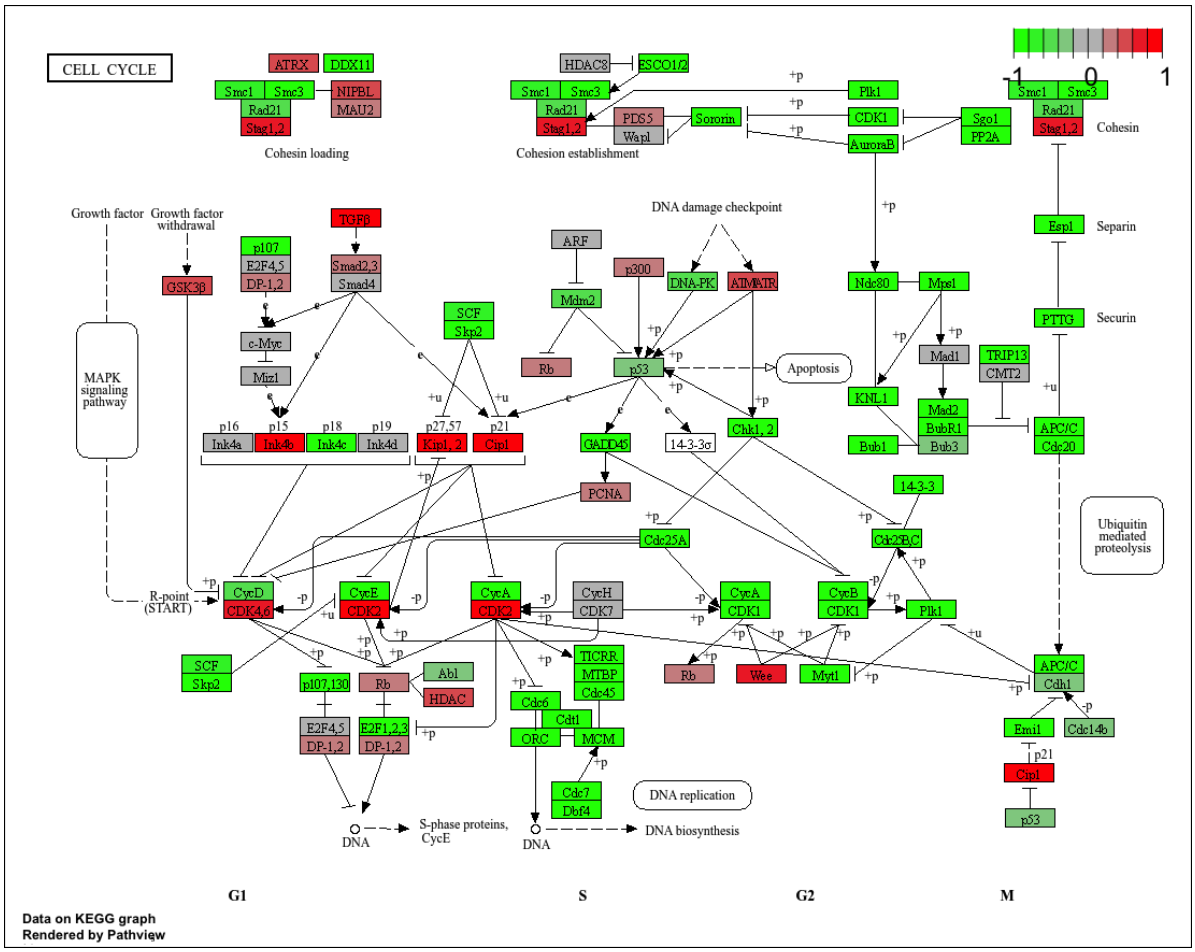


Figure 1: Cell cycle pathway colored by fold change

Top 5 Upregulated Pathways

```
keggrespathways <- rownames(keggres$greater)[1:5]
keggresids <- substr(keggrespathways, start = 1, stop = 8)
keggresids
```

```
[1] "hsa04640" "hsa04630" "hsa04142" "hsa00140" "hsa04740"
```

```
pathview(gene.data = foldchanges, pathway.id = keggresids, species = "hsa")
```

```
'select()' returned 1:1 mapping between keys and columns
```

```
Info: Working in directory /Users/abbyperez/Library/Mobile Documents/com~apple~CloudDocs/BIM

Info: Writing image file hsa04640.pathview.png

'select()' returned 1:1 mapping between keys and columns

Info: Working in directory /Users/abbyperez/Library/Mobile Documents/com~apple~CloudDocs/BIM

Info: Writing image file hsa04630.pathview.png

'select()' returned 1:1 mapping between keys and columns

Info: Working in directory /Users/abbyperez/Library/Mobile Documents/com~apple~CloudDocs/BIM

Info: Writing image file hsa04142.pathview.png

'select()' returned 1:1 mapping between keys and columns

Info: Working in directory /Users/abbyperez/Library/Mobile Documents/com~apple~CloudDocs/BIM

Info: Writing image file hsa00140.pathview.png

'select()' returned 1:1 mapping between keys and columns

Info: Working in directory /Users/abbyperez/Library/Mobile Documents/com~apple~CloudDocs/BIM

Info: Writing image file hsa04740.pathview.png
```

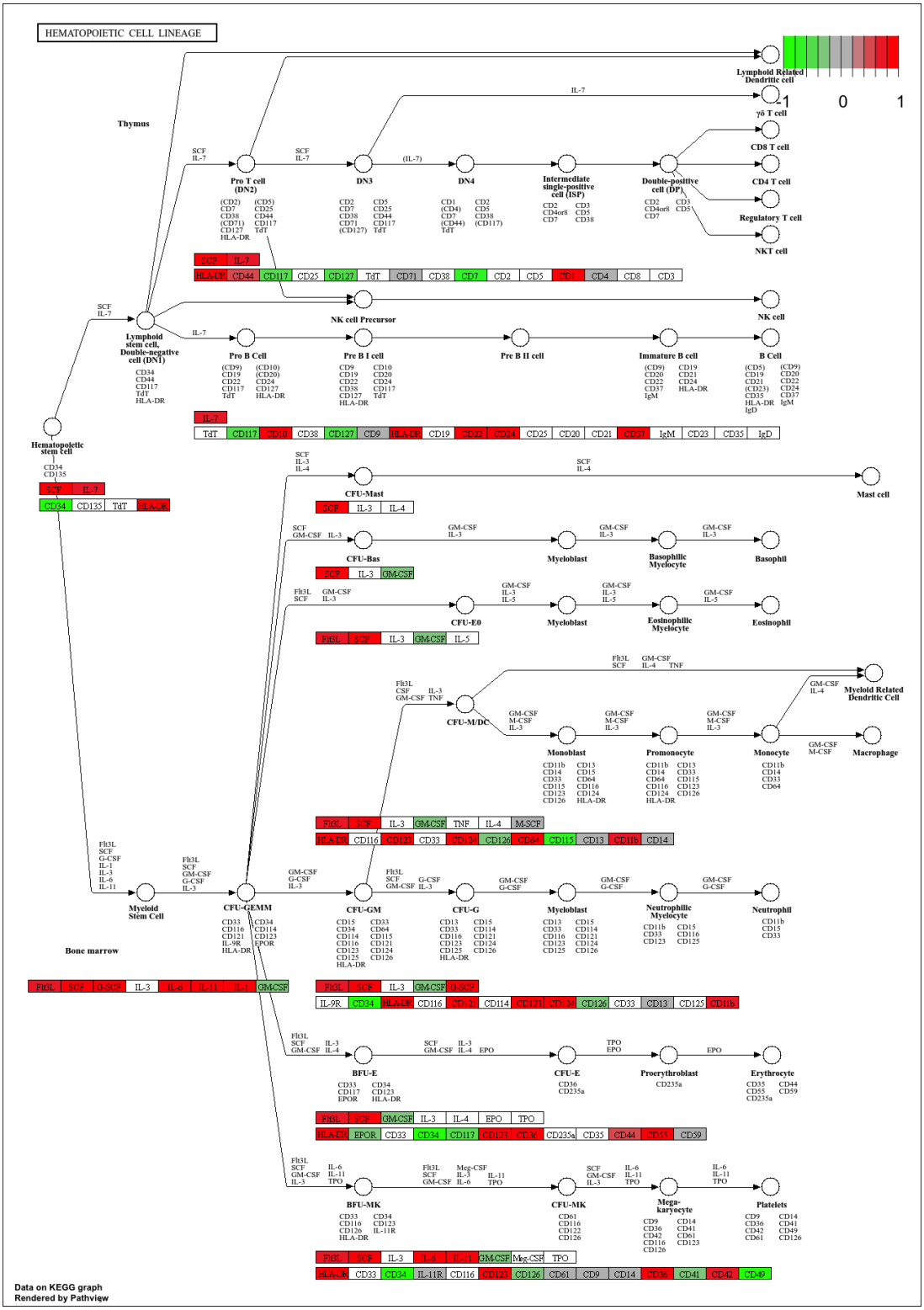


Figure 2: Hematopoietic cell lineage

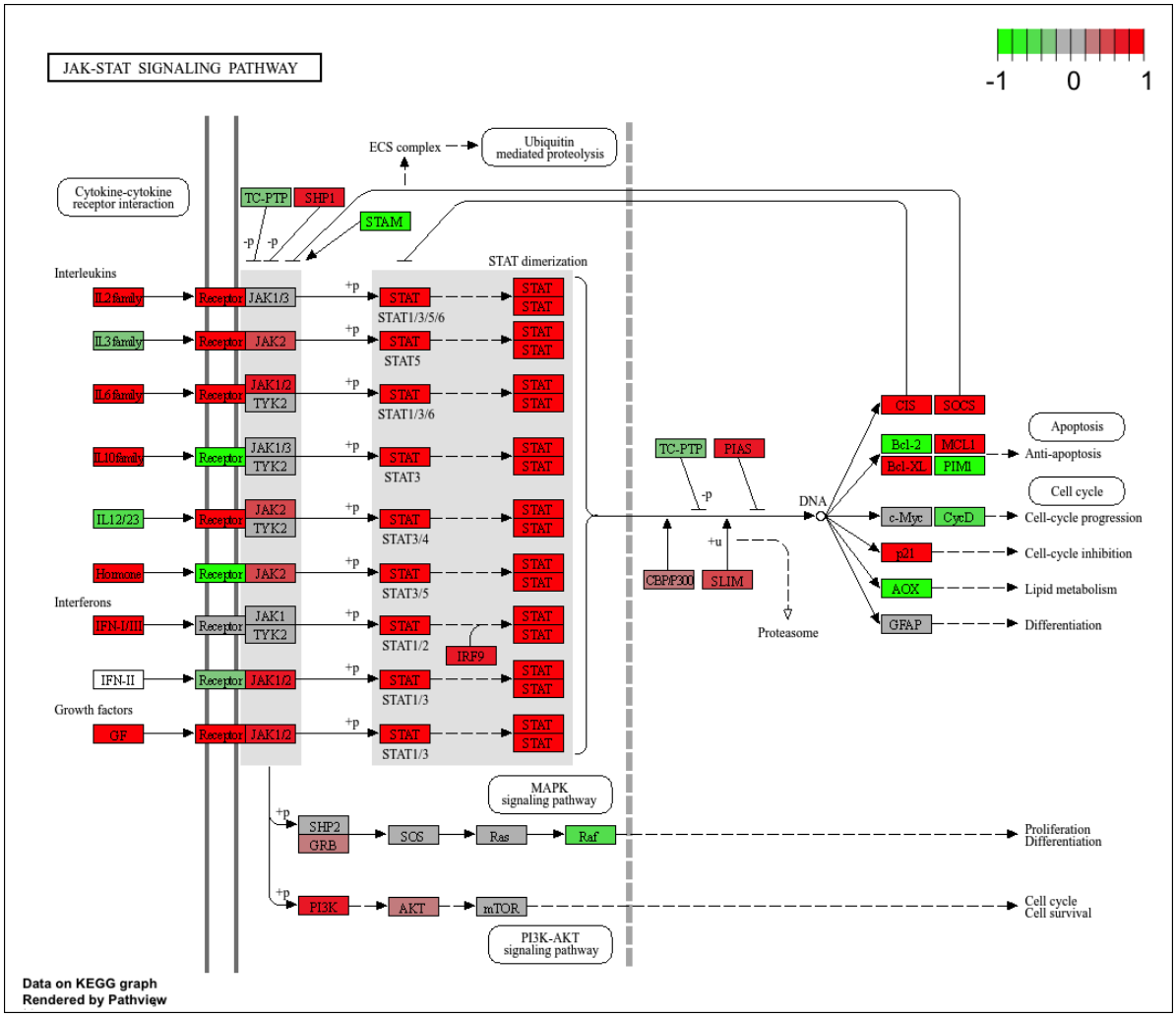


Figure 3: JAK-STAT signaling

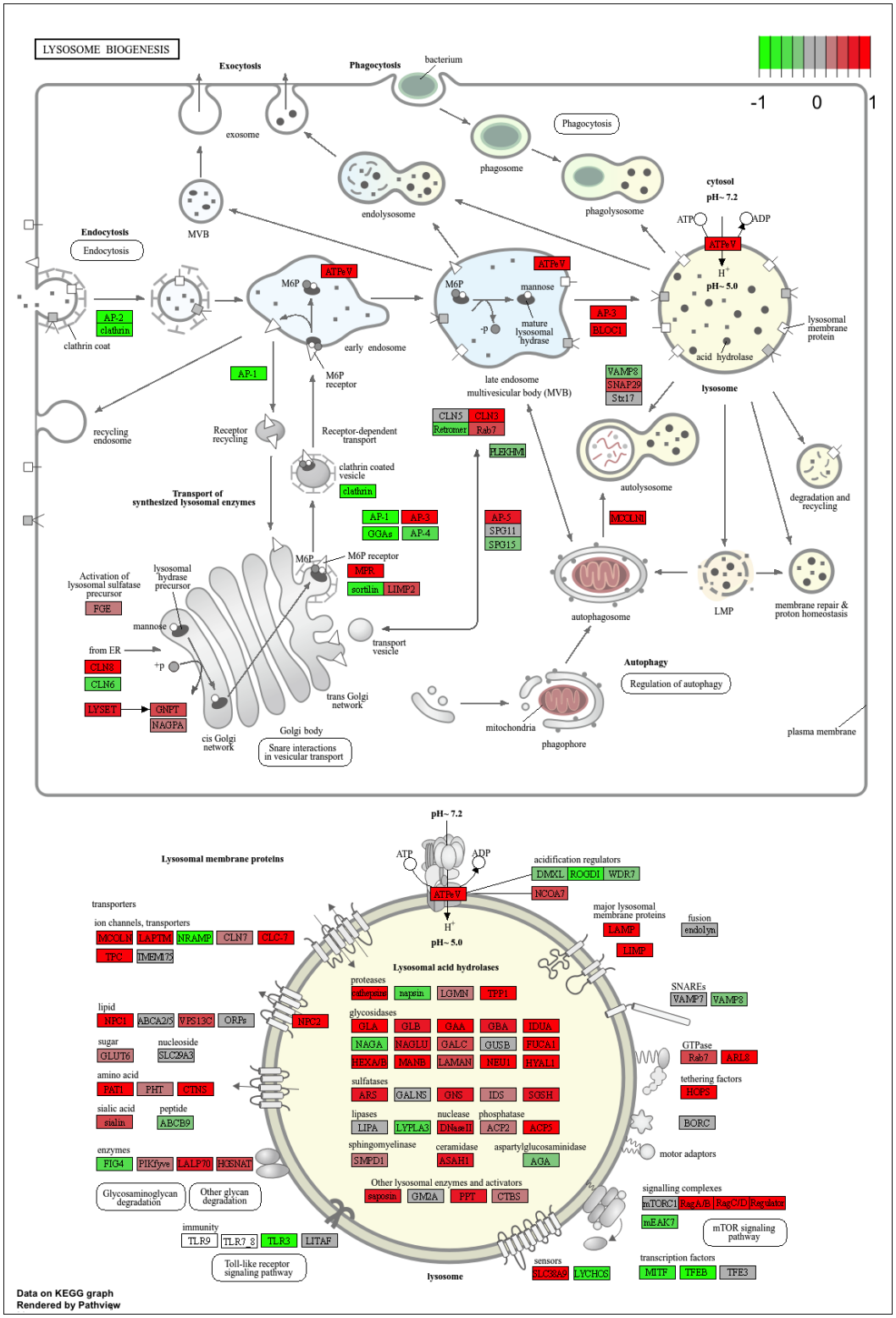


Figure 4: Lysosome

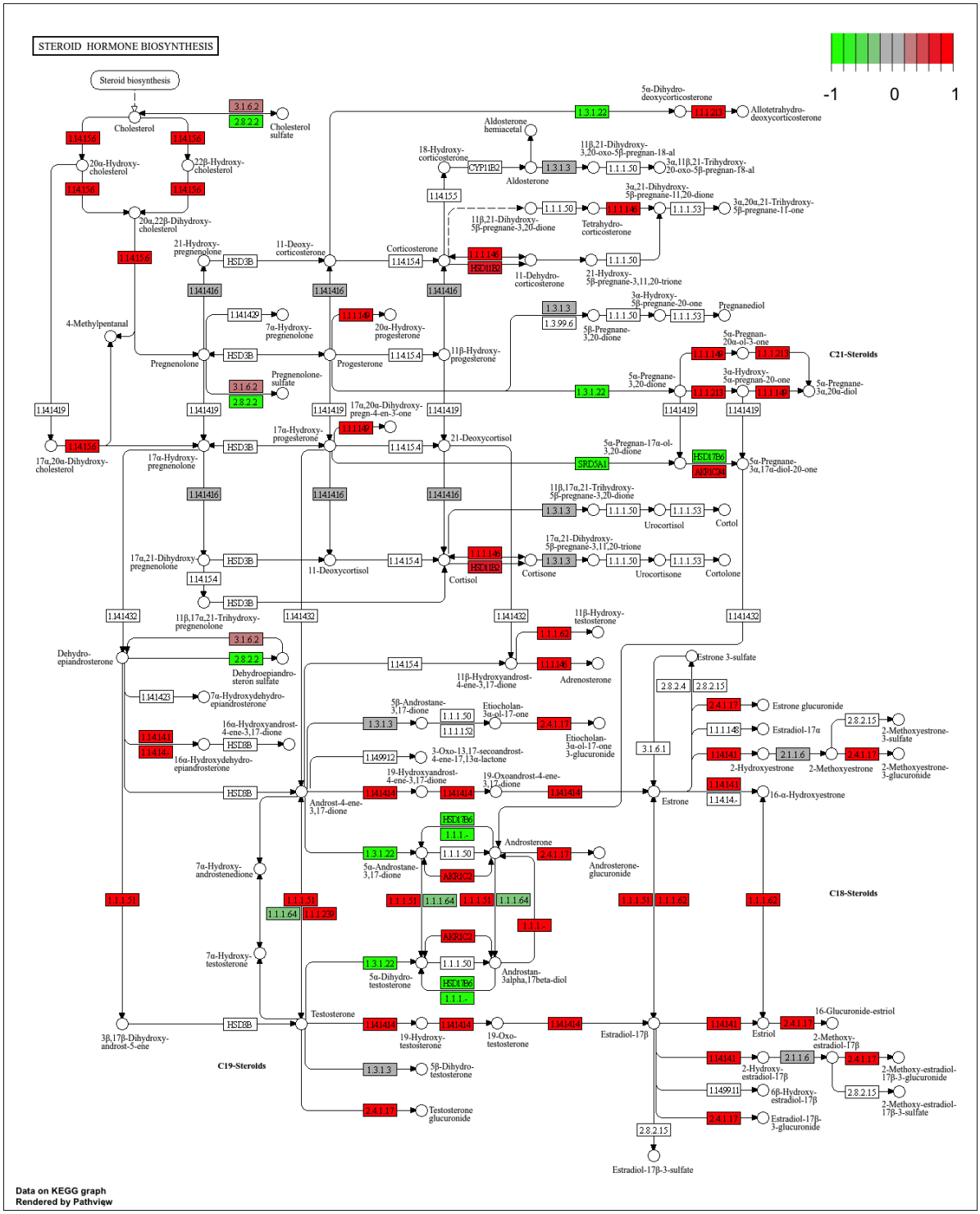


Figure 5: Steroid hormone biosynthesis

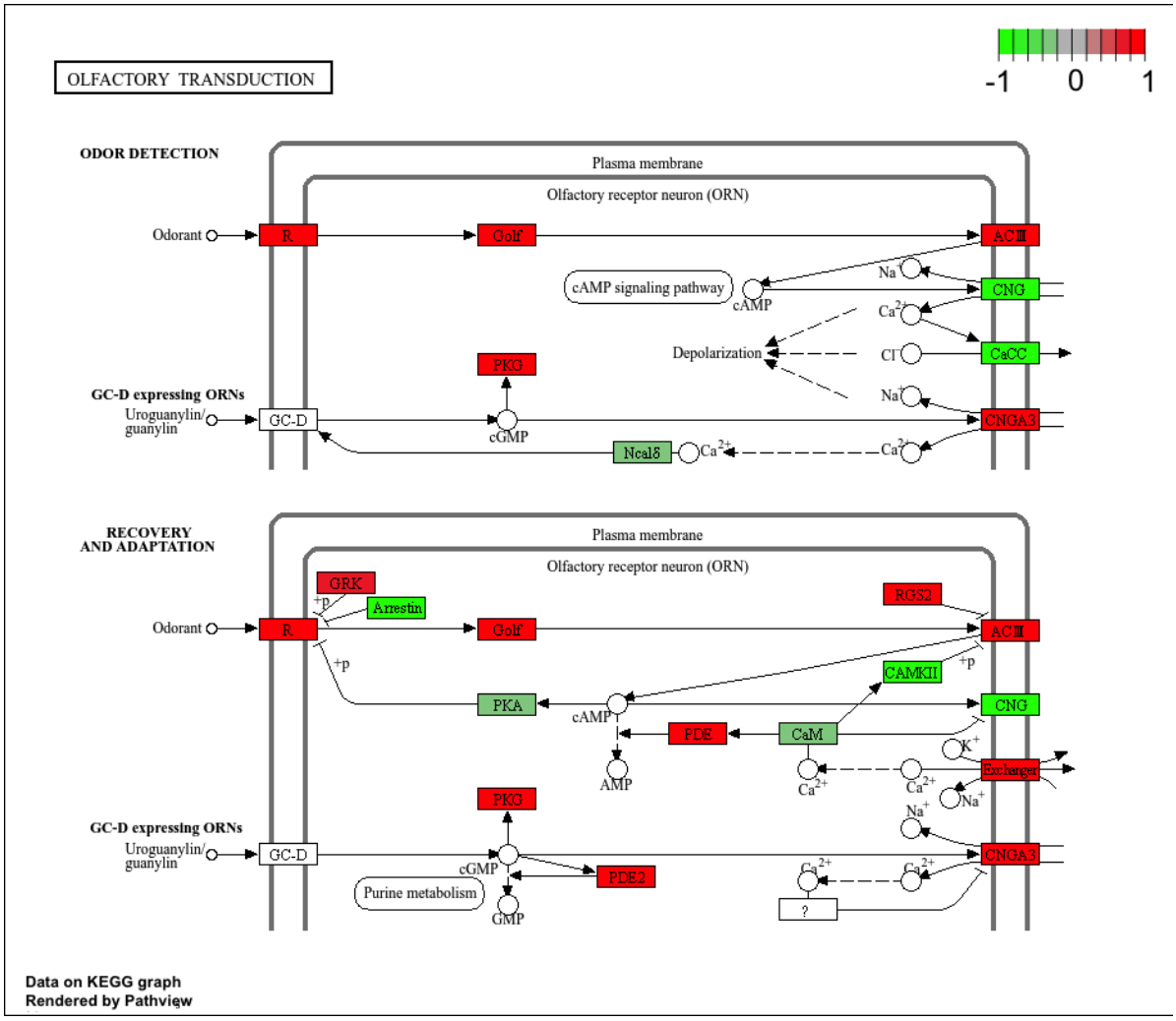


Figure 6: Olfactory transduction

Top 5 Downregulated Pathways

Q. Can you do the same procedure as above to plot the pathview figures for the top 5 down-regulated pathways?

```
keggrespathways_down <- rownames(keggres$less)[1:5]
keggresids_down <- substr(keggrespathways_down, start = 1, stop = 8)
keggresids_down
```

```
[1] "hsa04110" "hsa03030" "hsa03013" "hsa04114" "hsa03440"
```

```r
pathview(gene.data = foldchanges, pathway.id = keggresids_down, species = "hsa")
```

```
'select()' returned 1:1 mapping between keys and columns

Info: Working in directory /Users/abbyperez/Library/Mobile Documents/com~apple~CloudDocs/BIM

Info: Writing image file hsa04110.pathview.png

'select()' returned 1:1 mapping between keys and columns

Info: Working in directory /Users/abbyperez/Library/Mobile Documents/com~apple~CloudDocs/BIM

Info: Writing image file hsa03030.pathview.png

'select()' returned 1:1 mapping between keys and columns

Info: Working in directory /Users/abbyperez/Library/Mobile Documents/com~apple~CloudDocs/BIM

Info: Writing image file hsa03013.pathview.png

'select()' returned 1:1 mapping between keys and columns

Info: Working in directory /Users/abbyperez/Library/Mobile Documents/com~apple~CloudDocs/BIM

Info: Writing image file hsa04114.pathview.png

'select()' returned 1:1 mapping between keys and columns

Info: Working in directory /Users/abbyperez/Library/Mobile Documents/com~apple~CloudDocs/BIM

Info: Writing image file hsa03440.pathview.png
```

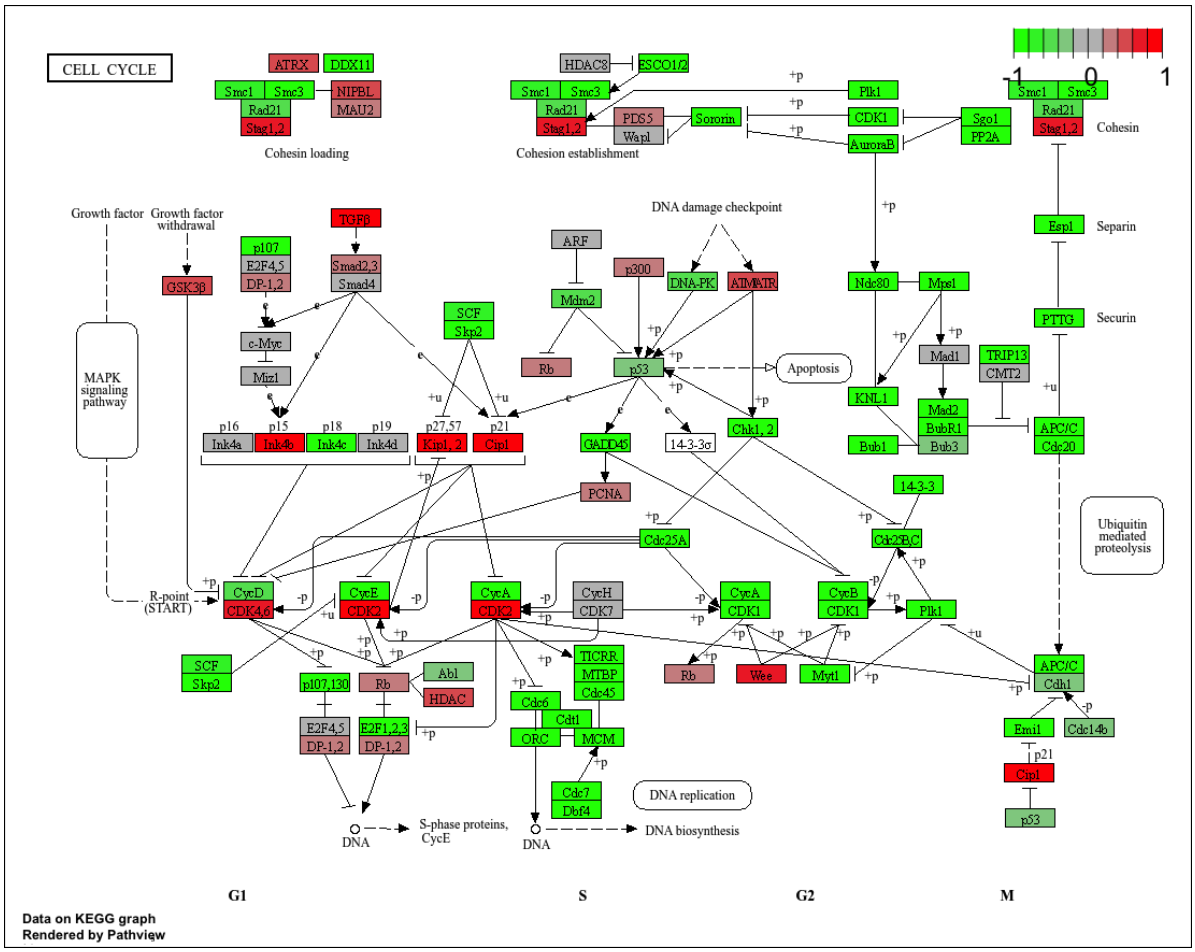


Figure 7: Cell cycle

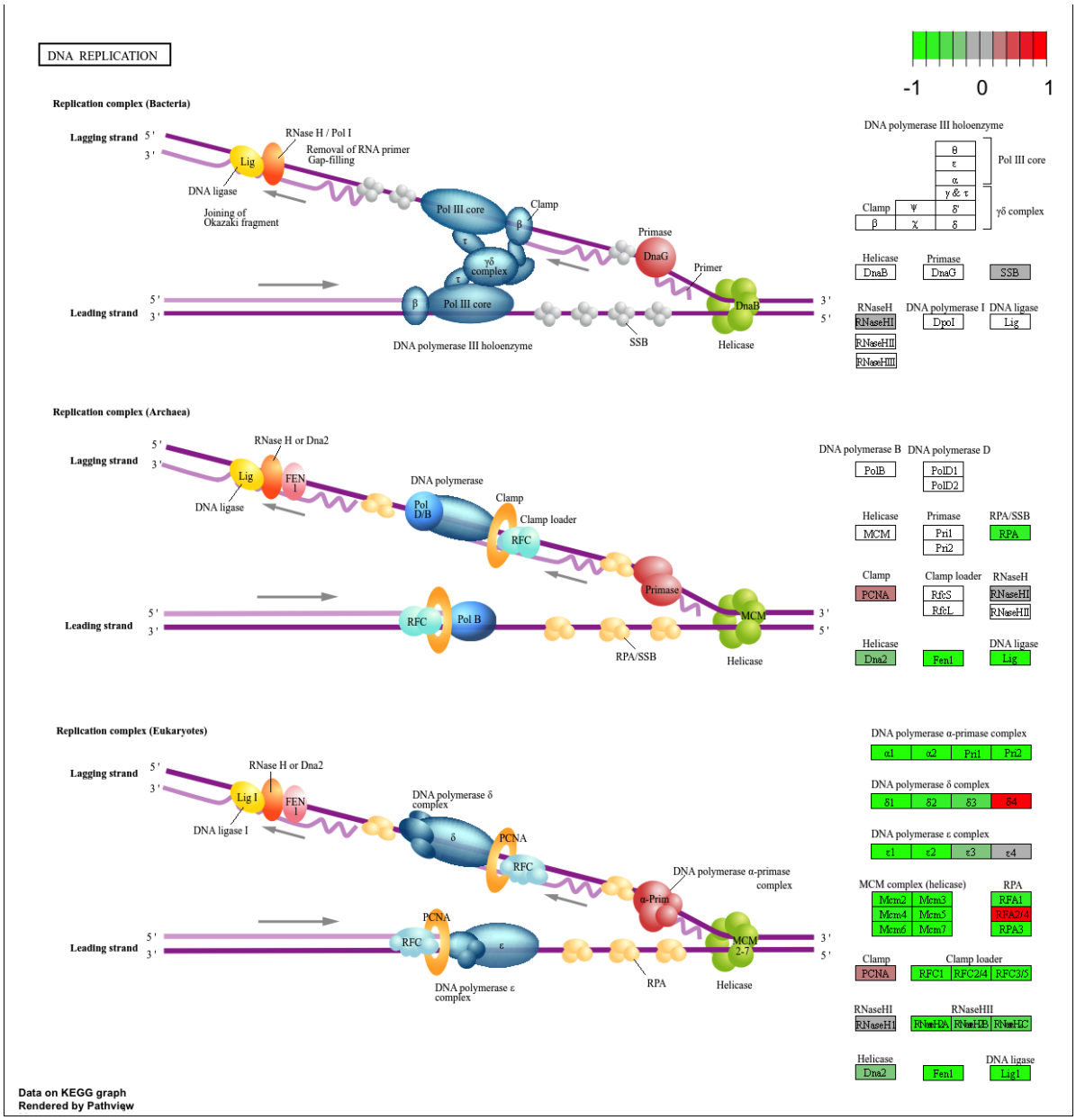


Figure 8: DNA replication

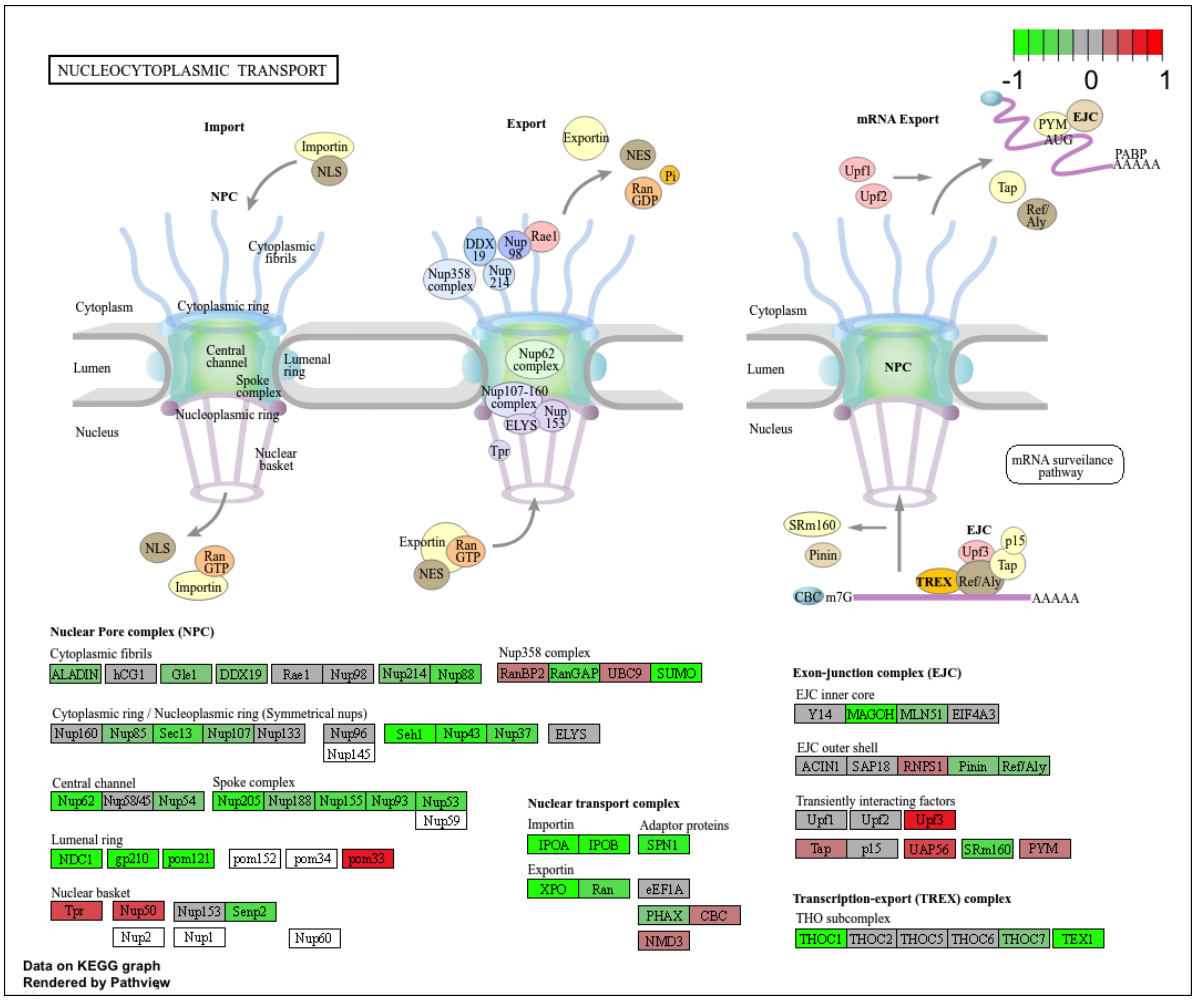


Figure 9: RNA transport

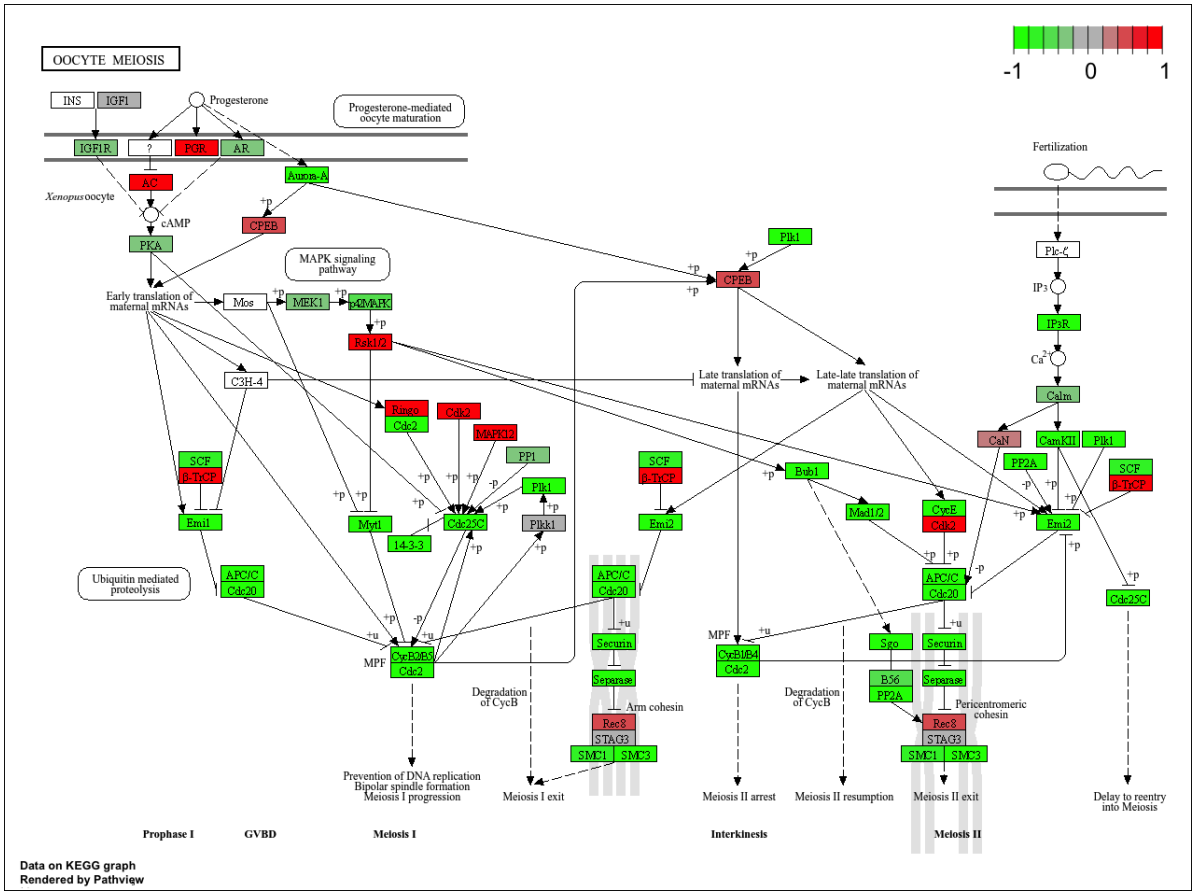


Figure 10: Oocyte meiosis

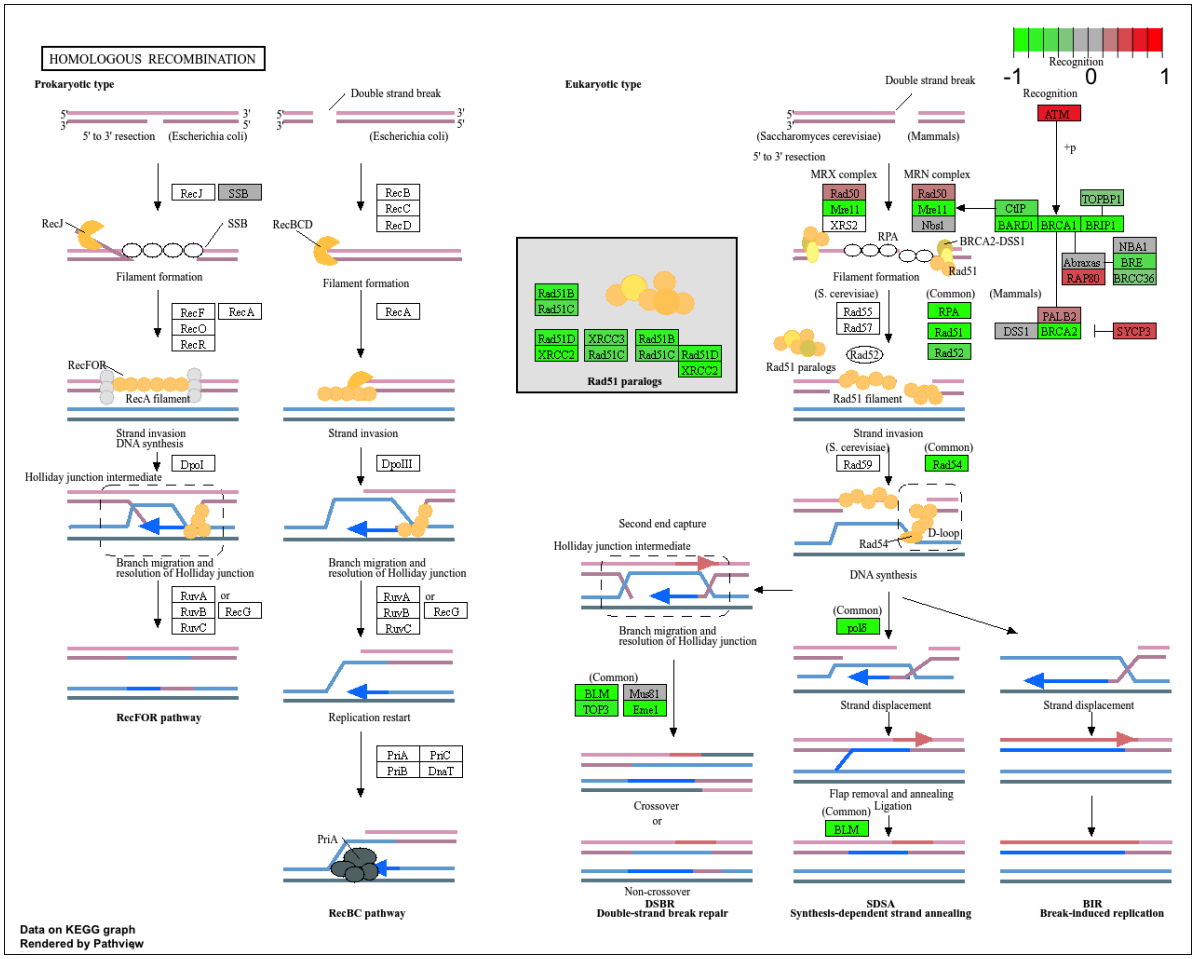


Figure 11: Homologous recombination

Section 3. Gene Ontology (GO)

```
data(go.sets.hs)
data(go.subs.hs)

# Focus on Biological Process
gobpsets <- go.sets.hs[go.subs.hs$BP]
gobpres <- gage(foldchanges, gsets = gobpsets)

lapply(gobpres, head)
```

```
$greater
                                            p.geomean stat.mean        p.val
GO:0007156 homophilic cell adhesion       4.892477e-05  3.971899 4.892477e-05
GO:0060429 epithelium development         6.727546e-05  3.834595 6.727546e-05
GO:0007610 behavior                       1.735208e-04  3.594057 1.735208e-04
GO:0048729 tissue morphogenesis           2.470962e-04  3.498983 2.470962e-04
GO:0002009 morphogenesis of an epithelium 3.227439e-04  3.429317 3.227439e-04
GO:0016337 cell-cell adhesion             8.195506e-04  3.163057 8.195506e-04
                                              q.val set.size         exp1
GO:0007156 homophilic cell adhesion       0.1337100      107 4.892477e-05
GO:0060429 epithelium development         0.1337100      478 6.727546e-05
GO:0007610 behavior                       0.2299150      402 1.735208e-04
GO:0048729 tissue morphogenesis           0.2455519      403 2.470962e-04
GO:0002009 morphogenesis of an epithelium 0.2565814      326 3.227439e-04
GO:0016337 cell-cell adhesion             0.3656453      318 8.195506e-04

$less
                                            p.geomean stat.mean        p.val
GO:0000279 M phase                        1.475361e-16 -8.323749 1.475361e-16
GO:0048285 organelle fission              7.498412e-16 -8.160305 7.498412e-16
GO:0000280 nuclear division               2.135098e-15 -8.034814 2.135098e-15
GO:0007067 mitosis                        2.135098e-15 -8.034814 2.135098e-15
GO:0000087 M phase of mitotic cell cycle 5.927567e-15 -7.891758 5.927567e-15
GO:0007059 chromosome segregation         1.055918e-11 -6.988373 1.055918e-11
                                               q.val set.size         exp1
GO:0000279 M phase                        5.864560e-13      492 1.475361e-16
GO:0048285 organelle fission              1.490309e-12      373 7.498412e-16
GO:0000280 nuclear division               2.121754e-12      349 2.135098e-15
GO:0007067 mitosis                        2.121754e-12      349 2.135098e-15
GO:0000087 M phase of mitotic cell cycle 4.712416e-12      359 5.927567e-15
GO:0007059 chromosome segregation         6.995458e-09      141 1.055918e-11

$stats
                                          stat.mean     exp1
GO:0007156 homophilic cell adhesion        3.971899 3.971899
GO:0060429 epithelium development          3.834595 3.834595
GO:0007610 behavior                        3.594057 3.594057
GO:0048729 tissue morphogenesis            3.498983 3.498983
GO:0002009 morphogenesis of an epithelium  3.429317 3.429317
GO:0016337 cell-cell adhesion              3.163057 3.163057
```

Section 4. Reactome Analysis

```
sig_genes <- res[res$padj <= 0.05 & !is.na(res$padj), "symbol"]
print(paste("Total number of significant genes:", length(sig_genes)))
```

```
[1] "Total number of significant genes: 8149"
```

```
write.table(sig_genes, file = "significant_genes.txt",
            row.names = FALSE, col.names = FALSE, quote = FALSE)
```

Upload `significant_genes.txt` to https://reactome.org/PathwayBrowser/#TOOL=AT, select "Project to Humans," and click Analyze.

Q. What pathway has the most significant Entities p-value? Do results match KEGG? What factors could cause differences?

The most significant pathway is typically a mitotic cell cycle or cell division process, which aligns with KEGG's top result of Cell cycle (hsa04110). Differences between methods arise because KEGG and Reactome use different pathway definitions and curation standards, so the same biological process may be split into sub-pathways or grouped differently depending on the database.

Section 5. GO online (OPTIONAL)

> **Q: What pathway has the most significant "Entities p-value"? Do the most significant pathways listed match your previous KEGG results? What factors could cause differences between the two methods?**

Session Information

```
sessionInfo()
```

```
R version 4.5.3 (2026-03-11)
Platform: aarch64-apple-darwin20
Running under: macOS Sonoma 14.8.2

Matrix products: default
BLAS:   /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/lib/libRblas.0.dylib
LAPACK: /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/lib/libRlapack.dylib;  R

locale:
[1] en_US.UTF-8/en_US.UTF-8/en_US.UTF-8/C/en_US.UTF-8/en_US.UTF-8

time zone: America/Los_Angeles
tzcode source: internal

attached base packages:
[1] stats4    stats     graphics  grDevices utils     datasets  methods
[8] base

other attached packages:
 [1] gageData_2.48.0             gage_2.60.0
 [3] pathview_1.50.0             org.Hs.eg.db_3.22.0
 [5] AnnotationDbi_1.72.0        ggplot2_4.0.3
 [7] DESeq2_1.50.2               SummarizedExperiment_1.40.0
 [9] Biobase_2.70.0              MatrixGenerics_1.22.0
[11] matrixStats_1.5.0           GenomicRanges_1.62.1
[13] Seqinfo_1.0.0               IRanges_2.44.0
[15] S4Vectors_0.48.1            BiocGenerics_0.56.0
[17] generics_0.1.4

loaded via a namespace (and not attached):
 [1] KEGGREST_1.50.0     gtable_0.3.6        xfun_0.57
 [4] lattice_0.22-9      bitops_1.0-9        vctrs_0.7.3
 [7] tools_4.5.3         parallel_4.5.3      tibble_3.3.1
[10] RSQLite_3.52.0      blob_1.3.0          pkgconfig_2.0.3
[13] Matrix_1.7-5        RColorBrewer_1.1-3  S7_0.2.2
[16] graph_1.88.1        lifecycle_1.0.5     compiler_4.5.3
[19] farver_2.1.2        Biostrings_2.78.0   codetools_0.2-20
[22] htmltools_0.5.9     RCurl_1.98-1.18     yaml_2.3.12
[25] GO.db_3.22.0        crayon_1.5.3        pillar_1.11.1
[28] BiocParallel_1.44.0 DelayedArray_0.36.1 cachem_1.1.0
[31] abind_1.4-8         tidyselect_1.2.1    locfit_1.5-9.12
[34] digest_0.6.39       dplyr_1.2.1         labeling_0.4.3
[37] fastmap_1.2.0       grid_4.5.3          cli_3.6.6
```

```
[40] SparseArray_1.10.10 magrittr_2.0.5      S4Arrays_1.10.1
[43] XML_3.99-0.23       withr_3.0.2         scales_1.4.0
[46] bit64_4.8.0         rmarkdown_2.31      XVector_0.50.0
[49] httr_1.4.8          bit_4.6.0           otel_0.2.0
[52] png_0.1-9           memoise_2.0.1       evaluate_1.0.5
[55] knitr_1.51          rlang_1.2.0         Rcpp_1.1.1-1.1
[58] glue_1.8.1          DBI_1.3.0           Rgraphviz_2.54.0
[61] KEGGgraph_1.70.0    rstudioapi_0.18.0   jsonlite_2.0.0
[64] R6_2.6.1
```